Lab 1 – Coaxial Cable

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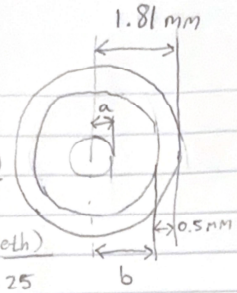
9/14/2025

Pre-Lab



$Z_0 = 50\,\Omega$

Conductor (PEC)

$\sigma_c = \infty \Rightarrow R_s = 0$

Dielectric (Polyeth)

$\sigma = 0 \quad \epsilon_r = 2.25$

$\mu = \mu_0$

$b = [1.81 - 0.5] = 1.31\,mm$

$R' = R_s\,[\sim\sim] = 0$

$G' = \sigma\,[\sim\sim] = 0$

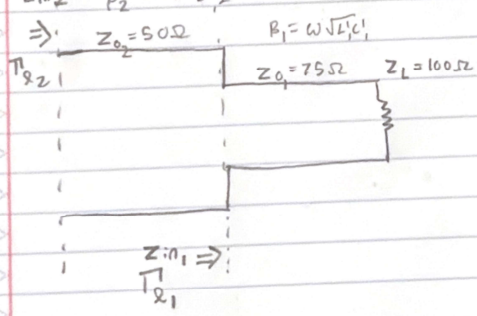
$R'C' = L'G' = 0 \therefore$ Lossless Line //

$Z_0 = \sqrt{\frac{L'}{C'}} \qquad L' = \frac{\mu}{2\pi}\ln(b/a) \qquad C' = \frac{2\pi\epsilon}{\ln(b/a)}$

$b/a = 2.42 \quad \boxed{a = 0.375\,mm}$ // Obs-1

Repeat for $Z_0 = 75\,\Omega$, $\boxed{a = 0.2\,mm}$ // Obs-2

Z_{in2} $\quad \beta_2 = \omega\sqrt{L'_2C'_2}$



$f = 3\,GHz \Rightarrow \omega = 6\pi E9, \quad \ell = 0.05\,m$

for lossless lines:

① $\beta_1 = \omega\sqrt{L'_1C'_1}$ ② $\Gamma_1 = \frac{Z_L - Z_{0_1}}{Z_L + Z_{0_1}}$

③ $\Gamma_{\ell_1} = \Gamma_1 e^{-2j\beta_1\ell}$

④ $Z_{in_1} = Z_0\frac{1+\Gamma_\ell}{1-\Gamma_\ell}$

Now let $Z_L = Z_{in_1}$ and Solve ① ~ ③ using β_2 and Z_{0_2} to find

$\Gamma_{\ell_2} = -0.0588 + j0.0004$

$20\log|\Gamma_{\ell_2}| = \boxed{-24.6\,dB}$

Obs-3 //

Part 1

The model shown in Figure 1 was used for all part 1 simulations. As designed in the pre-lab, it has an outer conductor with radius 1.31 mm, and an inner conductor with radius .375 mm, both made of copper, with a polyethylene dielectric layer filling the space in between. The coax section has a length of 5 cm.

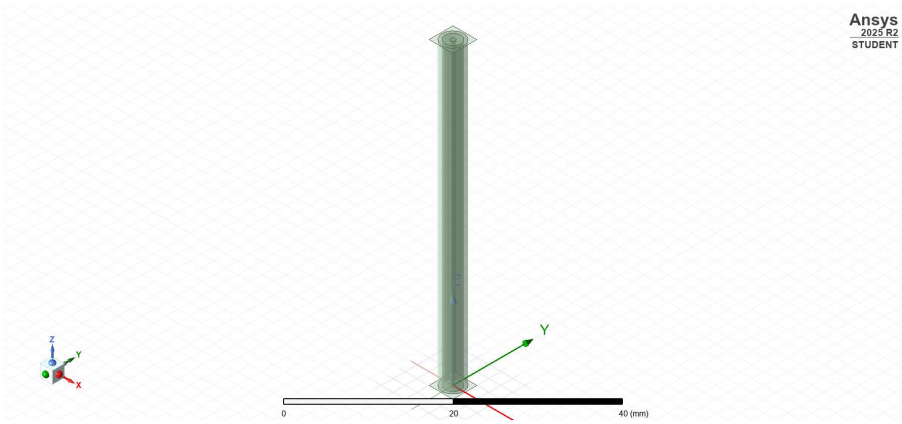


Figure 1: Device model for 50-Ohm coax

Modal Wave

The port setup for part A is shown in Figure 2 with Modal solution type and Wave ports.

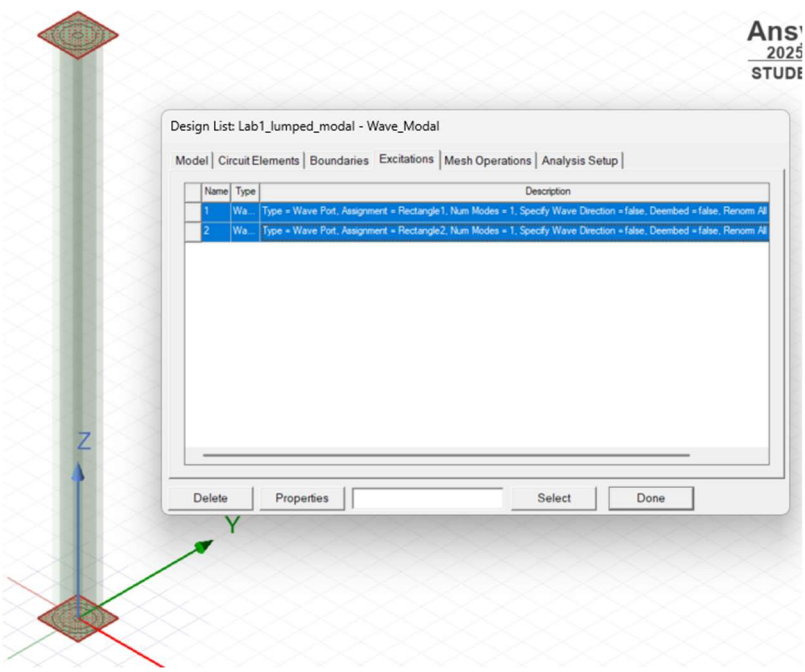


Figure 2: Port setup with Driven Modal and Wave Port

The simulated S-parameters for the 50-Ohm coax with Driven Modal and Wave Port are shown in Figures 3 and 4. Figure 3 shows the gain of port 1 relative to itself. The plot remains below -65 dB, showing that reflection remains negligible for all frequencies in the sweep. Figure 4 shows the gain of port 2 relative to

port 1. The graph's negative values indicate a small resistive loss in the line, which matches expectations simulating a realistic conductor. The negative slope indicates that this loss increases with frequency.

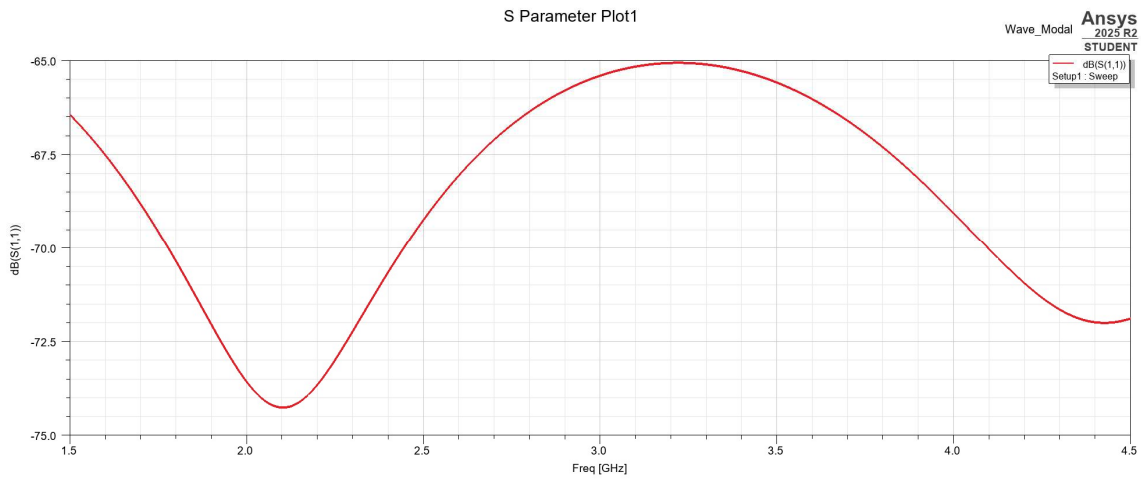


Figure 3: Plot of scattering parameter S(1,1) vs frequency with Driven Modal and Wave Port

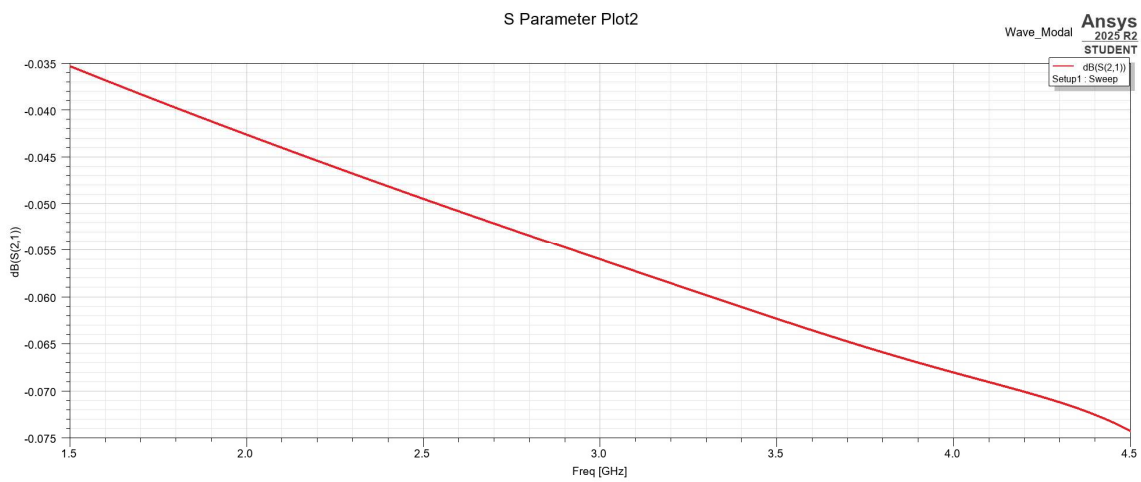


Figure 4: Plot of scattering parameter S(2,1) vs frequency with Driven Modal and Wave Port

The simulated characteristic impedance is shown in Figure 5. (Res-2) These results closely match the designed characteristic impedance of 50 Ohms. Discrepancies can be attributed to the assumption of PEC when the simulation used copper conductors.

Freq	Port Zo:1
2GHz	1 (50.106, -0.025845)
	2 (50.129, -0.025835)
3GHz	1 (50.097, -0.016481)
	2 (50.119, -0.016471)
4GHz	1 (50.091, -0.01088)
	2 (50.114, -0.01087)

Figure 5: Viewing Calculated Characteristic Impedance

Modal Lumped

The port setup for part B is shown in Figure 6 with Modal solution type and Lumped ports.

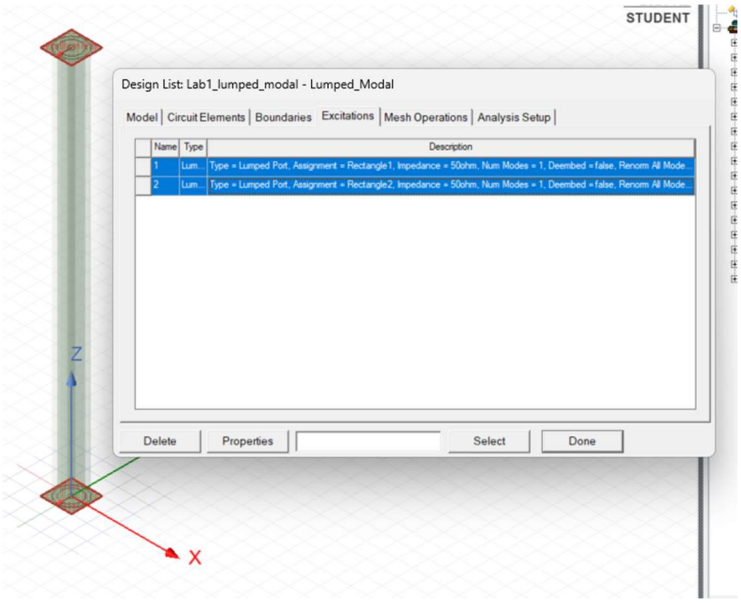


Figure 6: Port setup with Driven Modal and Wave Port

The simulated S-parameters for the 50-Ohm coax with Driven Modal and Lumped Port are shown in Figures 7 and 8. (Res-1) The plot for Figure 7 looks noticeably different from the Wave Port version, however reflection remains negligible for all frequencies in the sweep. Figure 8 is identical to the Wave Port version.

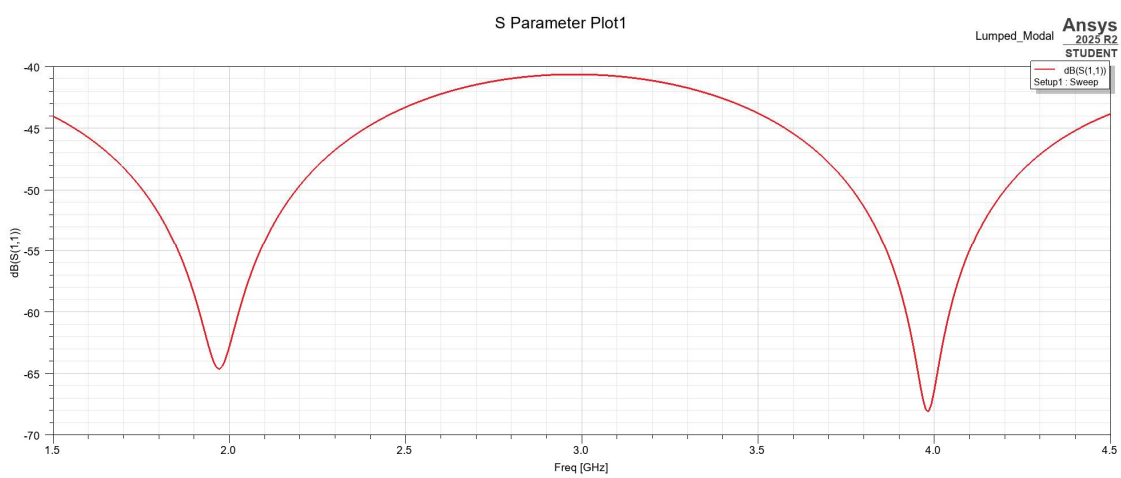


Figure 7: Plot of scattering parameter S(1,1) vs frequency with Driven Modal and Lumped Port

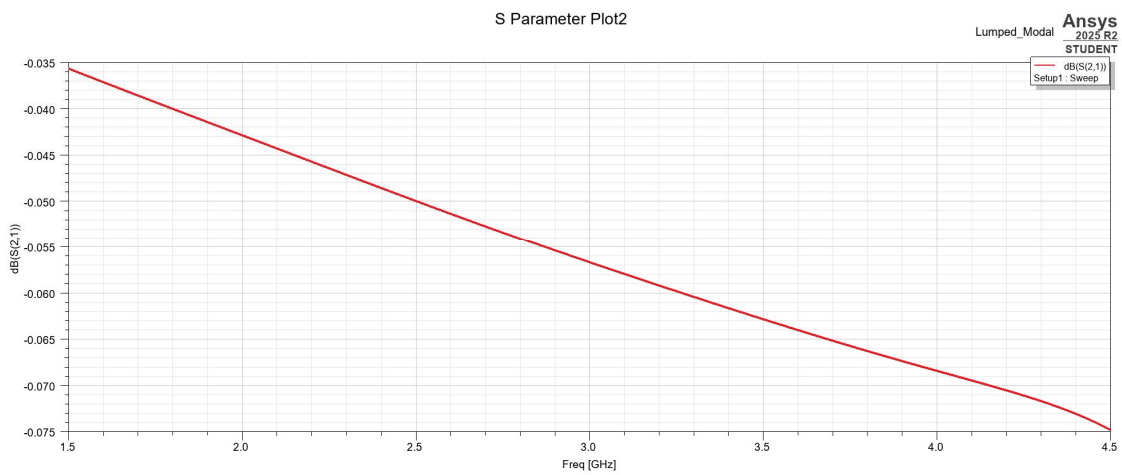


Figure 8: Plot of scattering parameter S(2,1) vs frequency with Driven Modal and Lumped Port

Terminal Wave

The model and port setup was unchanged from the Wave Modal case, but the solution type was changed to Terminal.

The simulated S-parameters for the 50-Ohm coax with Driven Terminal and Wave Port are shown in Figures 9 and 10. (Res-1) Figure 9 looks slightly different from the Modal version, however reflection remains negligible for all frequencies in the sweep. Figure 10 is identical to the Modal Wave and Lumped versions.

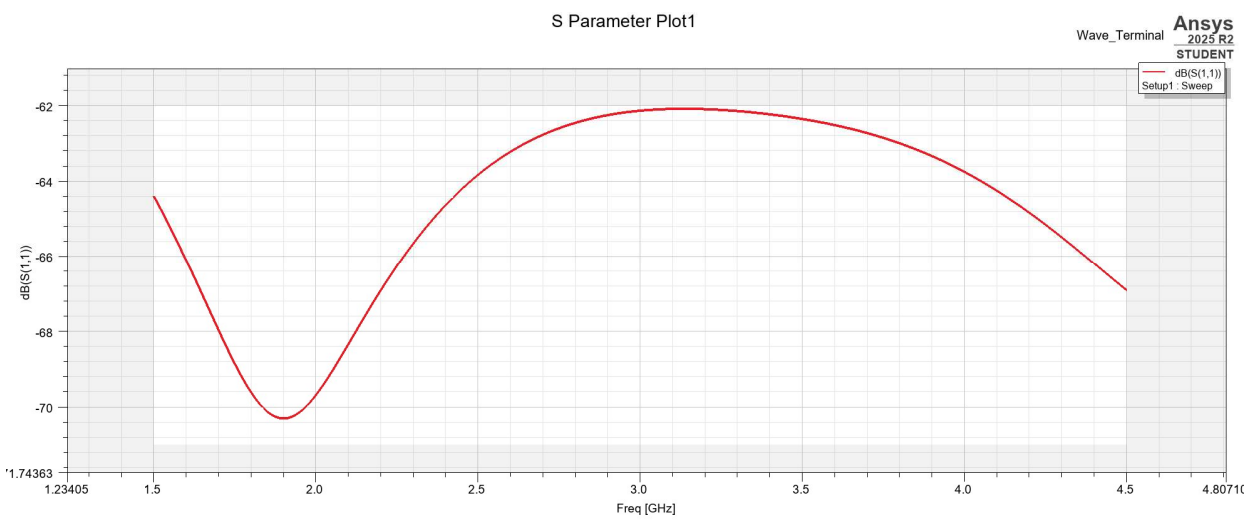


Figure 9: Plot of scattering parameter S(1,1) vs frequency with Driven Terminal and Wave Port

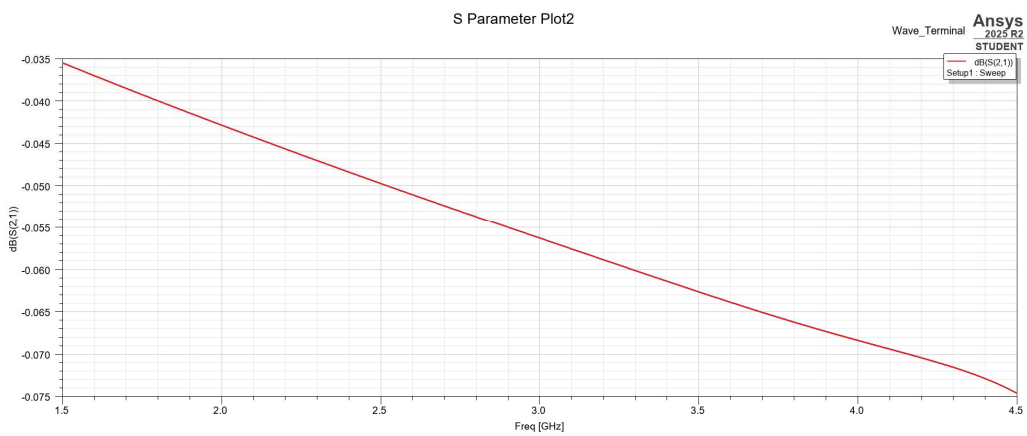


Figure 10: Plot of scattering parameter S(2,1) vs frequency with Driven Terminal and Wave Port

Terminal Lumped

The model and port setup was unchanged from the Lumped Modal case, but the solution type was changed to Terminal.

The simulated S-parameters for the 50-Ohm coax with Driven Terminal and Lumped Port are shown in Figures 11 and 12. (Res-1) Both plots are identical to their Modal counterparts.

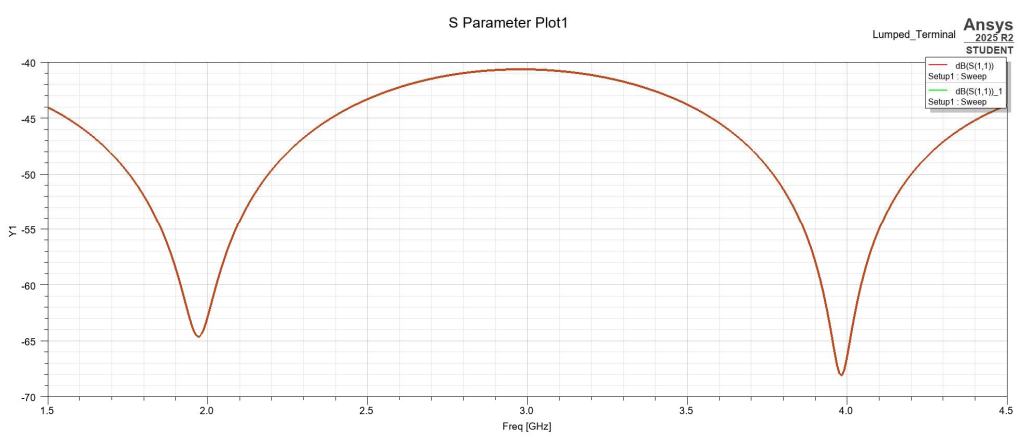


Figure 11: Plot of scattering parameter S(1,1) vs frequency with Driven Terminal and Lumped Port

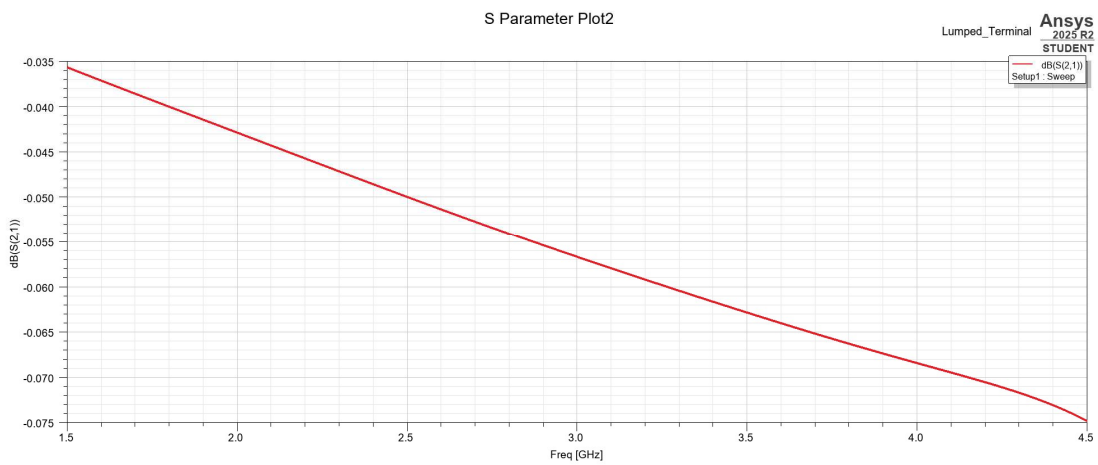


Figure 12: Plot of scattering parameter S(2,1) vs frequency with Driven Terminal and Lumped Port

Part 2

As shown in Figure 13, the test setup for part 2 consists of a 5-cm section of 50-Ohm coax followed by another 5-cm section of 75-Ohm coax.

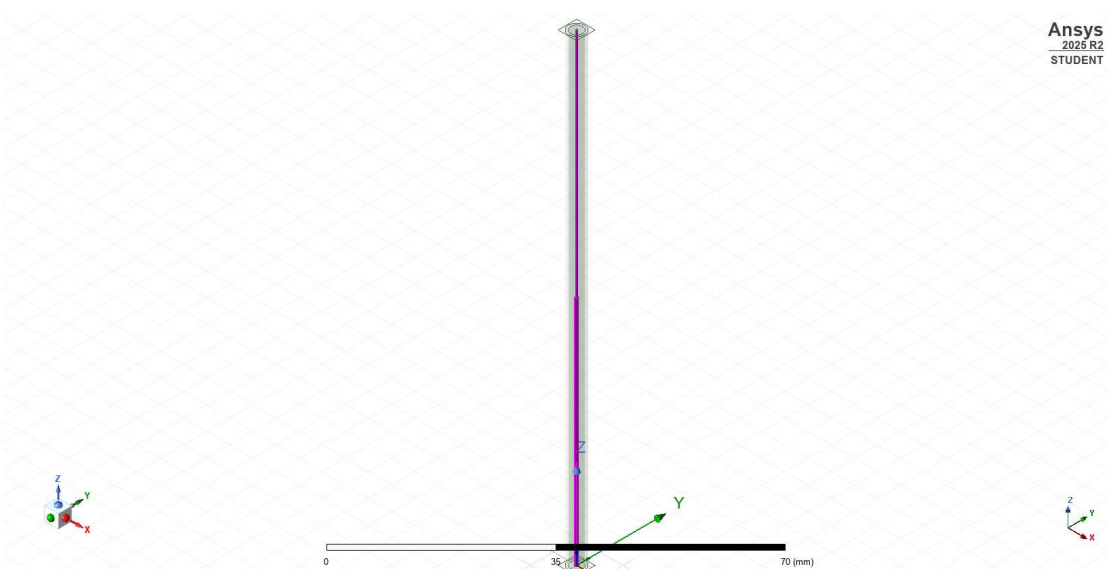


Figure 13: 50-Ohm coax followed by 75-Ohm coax.

For the simulation, the solution type is Driven Modal with Wave Ports. The port setup is shown in Figure 14. One port is set to renormalize as 100 ohms to represent the load, while the other is set to 50 Ohms to represent the infinite coax connected to the test segment.

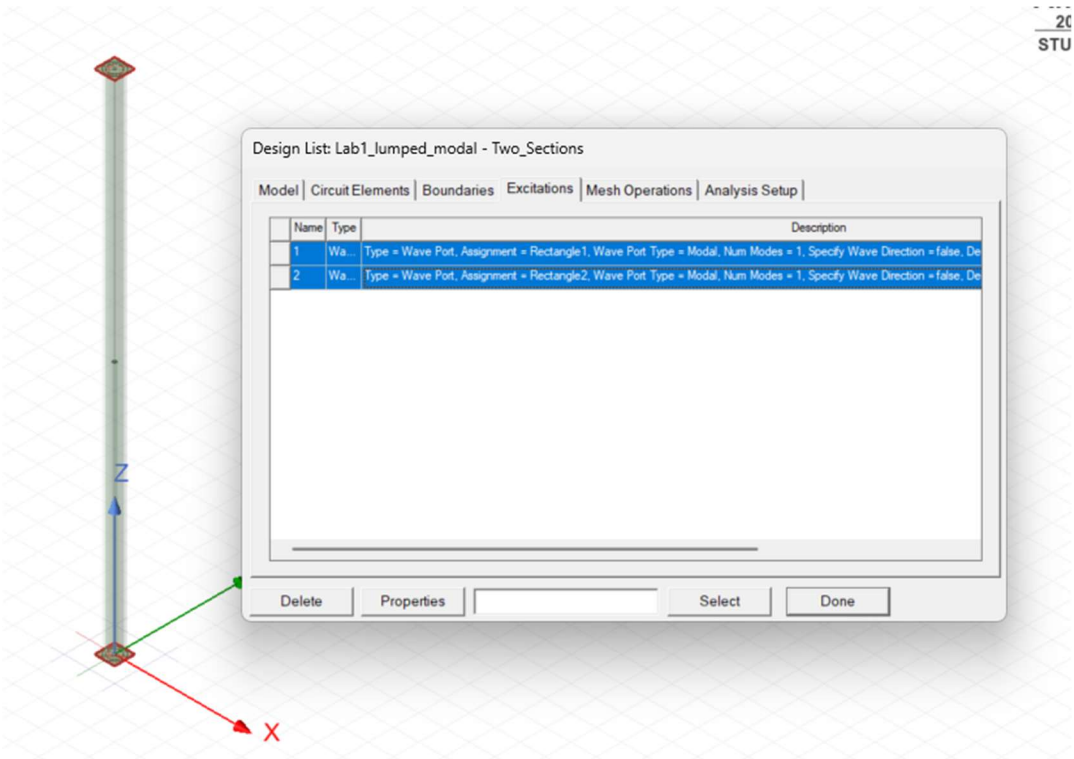


Figure 14: Port setup for two-section transmission line with renormalized Wave Ports.

Figures 15 and 16 show the plots of the scattering parameters for the model. (Res-3, Res-4) Figure 15 shows the reflection coefficient as -24 dB, which closely matches analytical predictions. The small discrepancy can be explained by the assumption of PEC when the simulation used copper conductors. Figure 16 shows higher average loss than the single sections, which is likely caused by the longer line as well as discontinuous characteristic impedance.

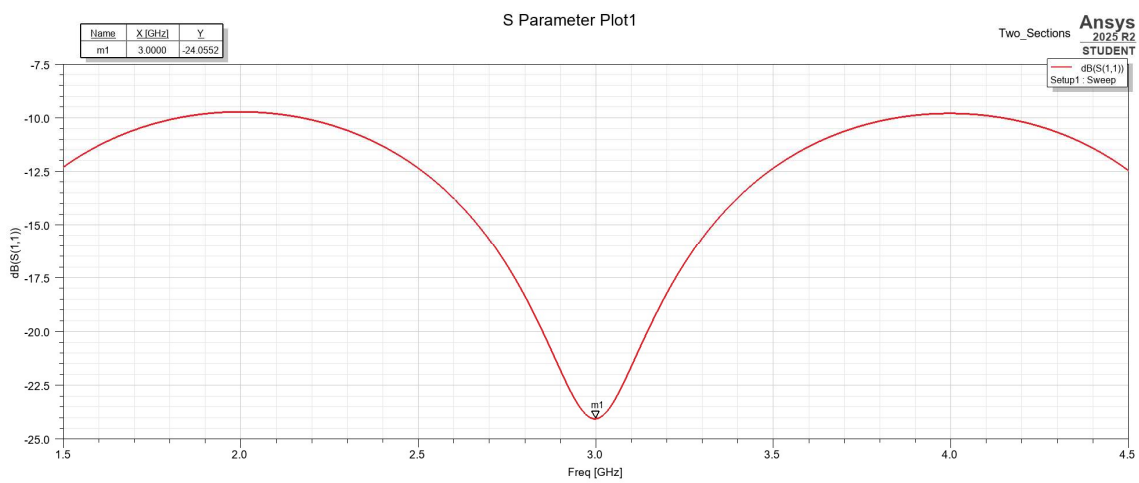


Figure 15: Plot of scattering parameter S(1,1) vs frequency for two-section coax with Driven Modal and Wave Port

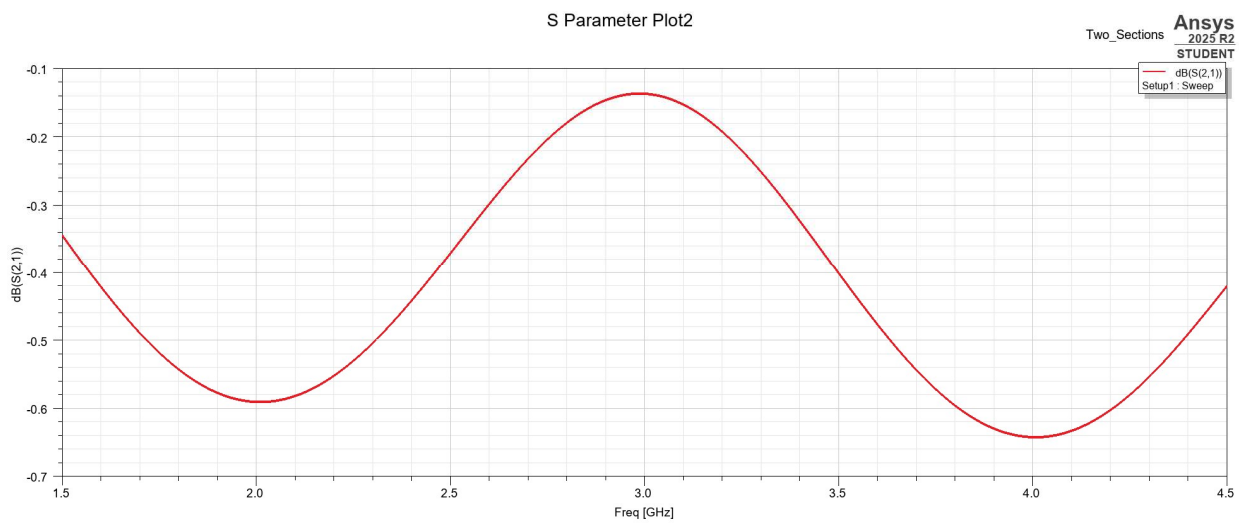


Figure 16: Plot of scattering parameter S(2,1) vs frequency for two-section coax with Driven Modal and Wave Port

1. I have included screenshots of all the HFSS models. (A.M.)
2. I have included screenshots of all port setups, including integration lines when appropriate (A. M.)
3. I have included all necessary results in graphs, tables, or statements (A.M.)
4. I have interpreted and commented on all my results (A.M.)